Notes: Bustos (AER, 2011)

Trade Liberalization, Exports, and Technology Upgrading: Evidence on the Impact of MERCOSUR on Argentinian Firms

Contents

1 Big picture

2 Data and measurement (key objects)

2.1 The trade policy experiment: MERCOSUR . . . . . .

2.2 Firm-level data . . . . . .

2.3 Technology spending . . . . . .

2.4 Binary innovation measures . . . . . .

2.5 Export status and entry . . . . . .

2.6 Productivity proxy: firm-size quartiles . . . . . .

2.7 Industry-level variables and controls . . . . . .

2.8 Key descriptive facts . . . . . .

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic environment . . . . . .

3.2 Profits under the four choices . . . . . .

3.3 Exit, export, and technology-adoption cutoffs . . . . . .

3.4 Sorting pattern of firms . . . . . .

3.5 Comparative statics of bilateral liberalization . . . . . .

3.6 Within-industry descriptive regression . . . . . .

3.7 Baseline export-entry specification . . . . . .

3.8 Alternative dynamic export specification . . . . . .

3.9 Export-entry heterogeneity by firm-size quartile . . . . . .

3.10 Technology-adoption specification . . . . . .

3.11 Technology heterogeneity by firm-size quartile . . . . . .

4 Identification and instruments (what makes the estimates credible)

4.1 Why Brazilian tariff cuts are a plausible source of exogenous variation . . . . . .

4.2 Why reverse causality is limited . . . . . .

4.3 Fixed effects and differencing . . . . . .

4.4 Two-digit industry trends . . . . . .

4.5 Industry characteristics that may shape tariff policy . . . . . .

4.6 Controlling for Argentina's own tariff changes . . . . . .

4.7 Sunk export costs and the bracketing argument . . . . . .

4.8 Sample-selection and measurement limitations . . . . . .

4.9 What the paper can and cannot distinguish . . . . . .

5 Tables and figures

5.1 Figure 1: Exporting and technology choices . . . . . .

5.2 Figure 2: How liberalization changes cutoffs . . . . . .

5.3 Table 1: Trade-policy and tariff facts . . . . . .

5.4 Table 2: Exporters versus nonexporters within industries . . . . . .

5.5 Table 3: Baseline effect on export entry . . . . . .

5.6 Table 4: Export entry by size quartile . . . . . .

5.7 Table 5: Average technology-spending response . . . . . .

5.8 Table 6: Binary innovation measures . . . . . .

5.9 Table 7: Technology adoption by size quartile . . . . . .

5.10 Tables 8 and 9: Mechanism by initial export status . . . . . .

5.11 Table 10: Industry-level exports to Brazil . . . . . .

5.12 Tables 11 and 12: Domestic sales . . . . . .

6 Short summary

7 Conclusion

7.1 Core conclusion (what the paper establishes) . . . . . .

7.2 Economic intuition . . . . . .

7.3 Takeaway . . . . . .

1 Big picture

- **Question.** Does trade liberalization raise firm performance only by reallocating market share toward already productive exporters, or can it also *induce* firms to upgrade technology?
- **Core contribution.** The paper adds endogenous technology adoption to a Melitz-style heterogeneous-firm trade model and then tests the model using the MERCOSUR liberalization between Argentina and Brazil.
- **Main mechanism.** Lower Brazilian tariffs raise export revenues for Argentinian firms. Because advanced technology has a fixed adoption cost but yields revenue-scaled benefits, larger export revenues can make technology upgrading profitable for firms that were previously below the adoption threshold.
- **Why this matters.** Earlier trade papers mostly emphasized selection and reallocation: productive firms export more, so aggregate productivity rises because those firms gain market share. Bustos shows that there is another channel: exporting opportunities can change firms' technology choices.
- **Setting.** The paper studies Argentina during the early MERCOSUR period. Brazil's tariffs on imports from Argentina fell sharply and differentially across industries, creating useful variation in export-market access.
- **Data advantage.** The firm survey observes *direct* measures of technology spending – computers, software, patents, technology transfers, R&D-related labor and materials, and innovation-related equipment – instead of inferring technology from productivity residuals.
- **Main empirical design.** The paper relates changes in export status and changes in technology spending from 1992 to 1996 to industry-level changes in Brazilian tariffs. Identification comes from cross-industry tariff variation combined with first differences and industry controls.
- **Main result on exporting.** Firms in industries with larger tariff cuts in Brazil were more likely to enter the export market. A 24 percentage-point average tariff reduction raised export entry by about 10–12 percentage points in the baseline specifications.
- **Main result on technology.** Firms in industries with larger tariff cuts increased technology spending faster. The average tariff reduction raised technology spending by about 0.24 log points in the baseline, with similar evidence using binary product and process innovation measures.
- **Most important heterogeneity result.** The effects are strongest in the *third quartile* of the initial firm-size distribution. This is exactly the upper-middle group predicted by the model: firms too productive to remain purely domestic, but not productive enough to have already exported and upgraded regardless of liberalization.
- **Mechanism evidence.** Technology upgrading appears for both firms that were already exporters in 1992 and firms that were not. This matters because it suggests the channel is increased revenues, not just the mechanical fact of becoming an exporter.
- **Supporting demand-side evidence.** At the industry level, lower Brazilian tariffs are associated with higher Argentine exports to Brazil. There is also suggestive evidence that lower Argentine tariffs against Brazil reduced domestic sales.

- **Main credibility point.** The results are robust to controls for Argentina's own tariff changes, industry characteristics that may shape tariff policy, two-digit industry trends, and alternative dynamic specifications for export entry.
- **Economic takeaway.** Trade liberalization can raise firm productivity not only by selecting better firms into exporting, but also by pushing firms already near the margin to adopt more advanced technology.

2 Data and measurement (key objects)

2.1 The trade policy experiment: MERCOSUR

The empirical setting is the bilateral liberalization between Argentina and Brazil under MERCOSUR. Brazil's most-favored-nation tariffs in 1991 averaged about 29 percentage points and were reduced to zero for Argentina by 1995. Over the paper's 1992–1996 window, the baseline measure of Brazilian tariff reductions is the 1992 tariff level, which averaged roughly 24 percentage points and varied widely across four-digit ISIC industries.

Argentina also changed its own tariffs during this period, but those movements were smaller and less central to the paper's mechanism. Argentina's tariffs against Brazil averaged about 13 percentage points initially, while unilateral tariffs against the rest of the world changed less because most unilateral liberalization had happened before 1992.

Interpretation. The key treatment is not "trade liberalization" in the abstract. It is the cross-industry variation in how much easier it became for Argentinian firms to sell into Brazil.

2.2 Firm-level data

The core micro data come from the *Encuesta Nacional de Innovación y Conducta Tecnológica de las Empresas Argentinas* (ENIT), conducted by Argentina's national statistics agency INDEC in 1997. The survey covers 1992–1996 and samples 1,639 industrial firms.

The sample is representative of firms with more than 10 employees. Even though only around 15% of establishments exceeded that size threshold, they accounted for over 90% of industrial output and value added, nearly 88% of employment, and over 94% of the wage bill.

For the main analysis, Bustos keeps a balanced panel of **1,380 firms** present in both 1992 and 1996 and belonging to four-digit ISIC industries for which Brazilian tariff data are available.

2.3 Technology spending

The paper's main technology variable is spending on technology, denoted ST. It includes:

- computers and software,
- payments for technology transfers and patents,
- spending on equipment, materials, and labor associated with innovation activities within the firm.

This measure is observed for every year from 1992 to 1996. That annual structure is valuable because it lets the paper collapse the panel into pre- and post-liberalization periods and difference out fixed firm heterogeneity.

Interpretation. ST is not a noisy residual from a production function. It is the firm's observed effort to upgrade technology.

2.4 Binary innovation measures

The survey also asks nine yes/no questions about whether the firm introduced particular improvements in products or production processes between 1992 and 1996. Bustos uses these responses to construct:

- an overall innovation index,
- a product innovation index,
- a process innovation index.

Each index is the fraction of relevant questions answered positively.

Interpretation. These measures are broader and available for a more representative set of firms than ST, especially at the bottom of the size distribution. They serve as robustness checks for the main technology-spending evidence.

2.5 Export status and entry

Export variables are available for 1992 and 1996. The paper classifies firms into:

- **continuing exporters**: export in both 1992 and 1996,
- **new exporters**: do not export in 1992 but export in 1996,
- **never exporters**: export in neither year,
- and a small group that exported in 1992 but not in 1996.

The main exporting outcome is either the change in export status between 1992 and 1996 or export status in 1996 controlling for export status in 1992.

2.6 Productivity proxy: firm-size quartiles

The theoretical model is about productivity thresholds, but the data do not provide the ingredients needed to estimate productivity in the usual production-function way. Bustos therefore proxies for initial productivity using firm size: log employment in efficiency units relative to the four-digit industry mean in 1992.

Firms are then sorted into quartiles of this relative size distribution.

Interpretation. The quartile analysis is meant to test whether tariff cuts matter most for firms around the export and technology-adoption thresholds.

2.7 Industry-level variables and controls

The empirical analysis combines the ENIT survey with industry-level measures of:

- Brazil's tariffs against Argentina,
- Argentina's tariffs against Brazil,
- Argentina's tariffs against the rest of the world,
- input tariffs as well as output tariffs,
- average capital intensity and skill intensity using U.S. data,
- industry demand elasticity from Broda and Weinstein,
- exports from Argentina to Brazil from UN COMTRADE.

The tariff data come from UNCTAD-TRAINS and are aggregated to four-digit ISIC industries using product-level trade weights.

2.8 Key descriptive facts

The paper highlights several striking descriptive patterns:

- exports from Argentina to Brazil quadrupled between 1992 and 1996, while exports to the rest of the world rose only about 60%,
- continuing exporters in 1992 were larger, more skill-intensive, and spent more on technology per worker than nonexporters,
- new exporters did not look especially technology-intensive in 1992, but they increased technology spending faster between 1992 and 1996,
- the subsample with positive ST in 1992 and 1993–1996 contains only 894 firms and is tilted toward larger firms, whereas the binary innovation sample is closer to representative by size.

Interpretation. These descriptive facts already line up with the model's sorting logic before the causal tariff analysis begins.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic environment

The model is a two-country, one-sector, monopolistic-competition framework with heterogeneous firms. Firms differ in productivity ϕ, face a fixed export cost f_x, and can choose between a low-technology production mode l and a high-technology production mode h.

The two cost functions are:

$$TC^l(q,\phi) = f + \frac{q}{\phi}, \qquad TC^h(q,\phi) = \eta f + \frac{q}{\gamma\phi}, \tag{1}$$

with $\eta > 1$ and $\gamma > 1$.

Interpretation. Technology h is more expensive because it has a higher fixed cost, but it reduces marginal cost by raising effective productivity from ϕ to $\gamma\phi$.

3.2 Profits under the four choices

Given CES demand, firms face four possible operating choices:

$$\begin{aligned}
\pi_d^l(\phi) &= \frac{1}{\sigma}E(P\rho)^{\sigma-1}\phi^{\sigma-1} - f,\\
\pi_d^h(\phi) &= \frac{1}{\sigma}E(P\rho)^{\sigma-1}\phi^{\sigma-1}\gamma^{\sigma-1} - \eta f,\\
\pi_x^l(\phi) &= (1+\tau^{1-\sigma})\frac{1}{\sigma}E(P\rho)^{\sigma-1}\phi^{\sigma-1} - f - f_x,\\
\pi_x^h(\phi) &= (1+\tau^{1-\sigma})\frac{1}{\sigma}E(P\rho)^{\sigma-1}\phi^{\sigma-1}\gamma^{\sigma-1} - \eta f - f_x.
\end{aligned} \tag{2}$$

Interpretation. Exporting raises revenue through access to the foreign market but requires paying the fixed export cost. High technology raises revenue because lower marginal cost allows the firm to sell more, but it requires the additional fixed adoption cost.

3.3 Exit, export, and technology-adoption cutoffs

The least productive firms exit. The exit cutoff ϕ^* solves:

$$\pi_d^l(\phi^*) = 0. \tag{3}$$

The export cutoff is:

$$\phi_x = \phi^*\tau\left(\frac{f_x}{f}\right)^{1/(\sigma-1)}. \tag{4}$$

The technology-adoption cutoff is defined by the indifference condition of the marginal exporter between low and high technology:

$$\pi_x^h(\phi_h) - \pi_x^l(\phi_h) = 0, \tag{5}$$

which implies

$$\phi_h = \phi^*(1+\tau^{1-\sigma})^{-1/(\sigma-1)}\left(\frac{\eta-1}{\gamma^{\sigma-1}-1}\right)^{1/(\sigma-1)}. \tag{6}$$

Interpretation. Lower trade costs τ reduce both the export cutoff and the technology-adoption cutoff. More firms start exporting, and some exporters also find it worthwhile to adopt advanced technology.

3.4 Sorting pattern of firms

The paper focuses on the empirically relevant case $\phi_x < \phi_h$. Then firms sort into four groups:

- $\phi < \phi^*$: exit,
- $\phi^* < \phi < \phi_x$: produce only for the domestic market using low technology,
- $\phi_x < \phi < \phi_h$: export but still use low technology,
- $\phi > \phi_h$: export and adopt high technology.

Interpretation. This ordering is crucial. It creates room for trade liberalization to induce entry into exporting for some firms and technology adoption for a subset of exporters that were previously just below the technology threshold.

3.5 Comparative statics of bilateral liberalization

When τ falls:

- the share of active firms that export rises,
- the share that adopt the high technology rises,
- the exit cutoff rises,
- the export and technology cutoffs fall.

The paper's key theoretical result is that trade liberalization can induce technology adoption because exporters' revenues rise more than domestic revenues fall. This result does *not* arise from a pure market-size effect. It relies on heterogeneous firms, fixed export costs, and fixed technology-adoption costs.

3.6 Within-industry descriptive regression

To compare exporters and nonexporters within the same four-digit industry, Bustos estimates regressions of the form:

$$\ln Y_{ij} = \alpha_1 NE_{ij} + \alpha_2 EE_{ij} + \alpha_3 EN_{ij} + I_j + \varepsilon_{ij}, \tag{7}$$

where NE are new exporters, EE are continuing exporters, EN are firms that exported in 1992 but not in 1996, and I_j are industry dummies.

Interpretation. This is not the causal design. It is a descriptive check of whether the model's sorting pattern appears in the raw cross-section.

3.7 Baseline export-entry specification

The export decision is modeled as an index model:

$$EXP_{ijst} = 1\{\beta_{\tau_x}\tau^x_{jt} + \alpha_{st} + \mu_i + \varepsilon_{ijst} > 0\}, \tag{8}$$

where j indexes four-digit industries, s two-digit sectors, and $t \in \{1992, 1996\}$.

Because a nonlinear fixed-effects estimator would suffer from the incidental-parameters problem, Bustos estimates the linear probability version:

$$EXP_{ijst} = \beta_{\tau_x} \tau^x_{jt} + \alpha_{st} + \mu_i + \varepsilon_{ijst}. \tag{9}$$

Taking first differences yields:

$$\Delta EXP_{ijs} = \beta_{\tau_x} \Delta\tau^x_j + \Delta\alpha_s + \Delta\varepsilon_{ijs}. \tag{10}$$

The richer specification adds tariff, firm, and industry controls:

$$\Delta EXP_{ijs} = \beta_{\tau_x} \Delta\tau^x_j + \beta_{\tau_m} \Delta\tau^m_j + \beta_z z_{ij,1992} + \beta_c c_j + \Delta\alpha_s + \Delta\varepsilon_{ijs}. \tag{11}$$

Interpretation. The coefficient of interest is β_{τ_x}. Since tariff changes are negative, a negative coefficient means that bigger tariff cuts raise export entry.

3.8 Alternative dynamic export specification

To address the possibility of sunk export costs, the paper also estimates:

$$EXP_{ijs,1996} = \beta_{\tau_x} \Delta\tau^x_j + \delta EXP_{ij,1992} + \alpha_s + \varepsilon_{ijs,1996}. \tag{12}$$

Interpretation. This levels specification cannot absorb fixed heterogeneity, but it serves as a bracketing exercise. If the main first-difference estimate is too large because of dynamics, this alternative should come in smaller in absolute value.

3.9 Export-entry heterogeneity by firm-size quartile

To test whether trade liberalization matters most for firms near the export threshold, Bustos estimates:

$$\Delta EXP_{ijs} = \sum_{r=1}^{4} \beta^r_{\tau_x} (\Delta\tau^x_j \times Q^r_{ij}) + \sum_{r=2}^{4} \delta_r Q^r_{ij} + \Delta\alpha_s + \Delta\varepsilon_{ijs}, \tag{13}$$

with richer versions adding the tariff and industry controls from equation (11).

Interpretation. This specification is a direct empirical translation of the threshold logic in the theory.

3.10 Technology-adoption specification

The level equation for technology spending is:

$$\log ST_{ijst} = \beta_{\tau_x} \tau^x_{jt} + \beta_{\tau_m} \tau^m_{jt} + \alpha_{st} + \mu_i + \varepsilon_{ijst}. \tag{14}$$

Because tariffs follow a predetermined liberalization path, estimating the yearly panel in levels would mechanically induce serial correlation. Bustos therefore collapses the post period and estimates the

first-difference version:

$$\Delta \log ST_{ijs} = \beta_{\tau_x} \Delta \tau_j^x + \beta_{\tau_m} \Delta \tau_j^m + \Delta \alpha_s + \Delta \varepsilon_{ijs}, \tag{15}$$

where

$$\Delta \log ST_{ijs} = \frac{1}{4} \sum_{t=1993}^{1996} \log ST_{ijst} - \log ST_{ijs,1992}. \tag{16}$$

Interpretation. The post period is averaged over 1993–1996, which uses all the technology-spending information while avoiding the Bertrand-Duflo-Mullainathan serial-correlation problem in a multi-period DiD.

3.11 Technology heterogeneity by firm-size quartile

The quartile-interaction specification is:

$$\Delta \log ST_{ijs} = \sum_{r=1}^{4} \beta_{\tau_x}^r (\Delta \tau_j^x \times Q_{ij}^r) + \sum_{r=2}^{4} \delta_r Q_{ij}^r + \beta_{\tau_m} \Delta \tau_j^m + \Delta \alpha_s + \Delta \varepsilon_{ij}. \tag{17}$$

The same equation is re-estimated using the overall innovation index and the product and process innovation subindexes as dependent variables.

Interpretation. This is the paper's sharpest empirical test of the theory. If the model is right, the third quartile – not the very smallest or very largest firms – should show the strongest response.

4 Identification and instruments (what makes the estimates credible)

4.1 Why Brazilian tariff cuts are a plausible source of exogenous variation

The paper's identification rests on two core facts.

- MERCOSUR tariff reductions were programmed in 1991 and reached zero for all industries by 1995.
- Brazil's 1991 MFN tariff structure applied to Argentina and the rest of the world alike, so it was not tailored to Argentina-specific industry shocks.

Since Argentina accounted for only 7.7% of Brazil's trade at the start of the period, it is hard to believe that Brazil's tariff schedule was designed around Argentina's firm-level technology decisions.

4.2 Why reverse causality is limited

The outcomes of interest are changes in export status and technology spending from 1992 to 1996. The tariff changes were largely predetermined by policy schedules laid out before these outcomes

were realized.

Interpretation. It is unlikely that firms upgraded technology first and somehow caused Brazil to lower tariffs in their industries.

4.3 Fixed effects and differencing

The first-difference design removes time-invariant plant and industry heterogeneity. That matters because Brazilian tariffs are not randomly assigned across industries.

What gets differenced out:

- fixed firm productivity,
- fixed sectoral parameters such as export-cost conditions,
- persistent industry characteristics correlated with tariffs.

Interpretation. The coefficient is identified from differential changes across industries, not from permanent differences between, say, chemical firms and food firms.

4.4 Two-digit industry trends

A remaining concern is that broad sectors were on different trends during the reform years. To address this, Bustos includes two-digit ISIC industry dummies in the differenced equations.

Interpretation. The comparison is then within broad sectors. For example, dairy is compared to macaroni, not to pharmaceuticals.

4.5 Industry characteristics that may shape tariff policy

Brazil's tariff structure could still correlate with deeper industry features. Bustos therefore controls for:

- capital intensity,
- skill intensity,
- demand elasticity.

All of these are measured with U.S. data to avoid endogeneity with Argentine outcomes.

Interpretation. This is the paper's main defense against omitted-variable bias from industry structure.

4.6 Controlling for Argentina's own tariff changes

Because Argentina also changed its tariff structure during the same period, the regressions include:

- output tariffs with respect to the world,
- input tariffs with respect to the world,

- output tariffs with respect to Brazil,
- input tariffs with respect to Brazil.

Interpretation. This matters because technology adoption depends on both export opportunities and domestic-market conditions. Without these controls, Brazilian tariff cuts might proxy for simultaneous shifts in Argentina's competitive environment.

4.7 Sunk export costs and the bracketing argument

If export entry has persistence because of sunk costs, then a pure first-difference model may overstate the effect of tariff cuts. Bustos therefore estimates the lagged-dependent-variable specification in levels and argues that the two approaches bracket the truth.

The levels estimates are indeed somewhat smaller in absolute value, but remain economically large and statistically significant.

Interpretation. The export-entry result is not an artifact of ignoring export persistence.

4.8 Sample-selection and measurement limitations

The paper is transparent about several limits.

- The balanced panel excludes firms that exited after 1992.
- The main *ST* regressions use only firms with positive technology spending in 1992 and in the post period, which overrepresents larger firms.
- Firm size is only a proxy for productivity, so quartile results should not be interpreted too mechanically.

These limitations explain why the quartile patterns are strongest in the middle but not perfectly sharp in every other quartile.

4.9 What the paper can and cannot distinguish

The paper can show that improved export access induced technology upgrading. It is much less able to say whether the mechanism is exactly process innovation, product innovation, or some mixture of both – though the binary innovation results suggest both move. It can also show that technology upgrading appears in both old and new exporters, which favors a revenue channel, but it cannot observe the full internal decision process inside the firm.

5 Tables and figures

5.1 Figure 1: Exporting and technology choices

Read. Figure 1 is the theory picture. It plots four profit schedules against transformed productivity: low-tech domestic, high-tech domestic, low-tech export, and high-tech export.

Economic message. The figure makes the whole paper easy to remember. There are two thresholds above the exit cutoff: one for exporting and one for adopting advanced technology. Trade liberalization shifts these thresholds downward.

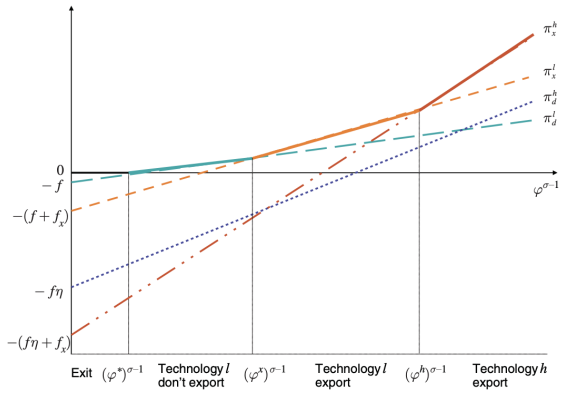

Figure 1. Exporting and Technology Choices

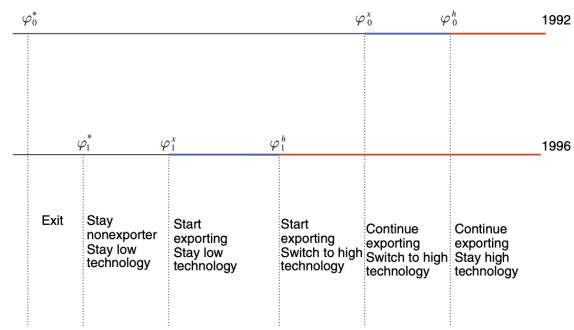

Figure 2. Effect of Falling Variable Trade Costs

5.2 Figure 2: How liberalization changes cutoffs

Read. Figure 2 is the conceptual empirical map. The pre-liberalization cutoffs ϕ_x^0 and ϕ_h^0 lie above the post-liberalization cutoffs ϕ_x^1 and ϕ_h^1.

Economic message. This is why the *upper-middle* firms matter. Some firms cross the export threshold, some cross the technology threshold, and some do both.

5.3 Table 1: Trade-policy and tariff facts

Read. Brazil's tariffs against Argentina were very high before MERCOSUR and then fell sharply. The average 1992 baseline tariff that matters for the 1992–1996 analysis is about 24 percentage points. Argentina's tariff changes against Brazil are much smaller, around 13 percentage points on average.

Economic message. The treatment intensity is large and varies substantially across industries. That is exactly what the identification needs.

Table 1—Brazil and Argentina's m.f.n. Tariffs

	Average	Standard deviation	Minimum	Maximum	Industries
Brazil's m.f.n. tariffs in 1991	0.29	0.17	0.00	0.84	101
Brazil's m.f.n. tariffs in 1992	0.24	0.13	0.00	0.63	104
Argentina's m.f.n. tariffs in 1992					
Outputs	0.13	0.06	0.00	0.22	102
Inputs	0.11	0.03	0.01	0.17	101
Change in Argentina's tariffs w.r.t. the world 1992–1996					
Outputs	0.01	0.05	−0.10	0.14	104
Inputs	0.01	0.02	−0.03	0.06	101

Note: Industries refer to 4-digit ISIC industries with available tariff data.

Table 2—Differences between Exporters and Nonexporters

	Levels in 1992		Changes 1992–1996		
	Continuing exporters	New exporters	Continuing exporters	New exporters	Number of firms
Firm characteristic					
Sales	1.770	1.032	0.180	0.254	1,380
	[0.089]***	[0.103]***	[0.040]***	[0.049]***	
Employment	1.509	0.870	0.021	0.183	1,380
	[0.074]***	[0.089]***	[0.026]	[0.035]***	
Spending on technology per worker	0.335	0.193	0.277	0.375	894
	[0.149]**	[0.175]	[0.106]***	[0.117]***	
Skill intensity	5.147	1.471	1.238	1.252	1,380
	[1.034]***	[1.104]	[0.377]***	[0.471]***	

Notes: Robust standard errors in brackets. Exporter premia are estimated from a regression of the form: $\ln Y_{ij} = \alpha_1 NE_{ij} + \alpha_2 EE_{ij} + \alpha_3 EN_{ij} + I_j + \varepsilon_{ij}$ where i indexes firms, j indexes four-digit ISIC industries; NE are new exporters (231 firms), EE are continuing exporters (556 firms), EN are firms that exported in 1992 but didn't in 1996 (27 firms) and the reference category relative to which differences are estimated is nonexporters (566 firms); I are industry dummies, and Y is the firm characteristic for which the differences are estimated.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.4 Table 2: Exporters versus nonexporters within industries

Read. Continuing exporters are much larger and more technology-intensive than never exporters in 1992. For example, continuing exporters have sales 1.770 log points above nonexporters, employment 1.509 log points higher, and technology spending per worker 0.335 log points higher. New exporters are also larger ex ante, but their 1992 technology-spending premium is smaller and less precise. Between 1992 and 1996 both continuing exporters and new exporters increase technology spending per worker faster than nonexporters.

Economic message. The within-industry sorting pattern already looks like the model: low-productivity firms stay domestic and low-tech, medium firms are the new export entrants, and top firms are established exporters with more advanced technology.

5.5 Table 3: Baseline effect on export entry

Read. In the first-difference baseline, the coefficient on Δ Brazil tariffs is about −0.421. With the full set of controls it stays in the −0.466 to −0.533 range. Since the average tariff decline is 24 percentage points, this implies a roughly 10–12 percentage-point increase in export entry. In the levels specification with lagged export status, the coefficients are somewhat smaller in magnitude but still large. In the sample of 1992 nonexporters, the implied effect is around 11–14.5 percentage points.

Economic message. Better access to Brazil caused firms to start exporting. This is not just selection on preexisting exporter characteristics.

Table 3—Entry in the Export Market

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Full sample. Dependent variable: change in export status 1996–1992								
Δ Brazil's tariffs	−0.421	−0.416	−0.407	−0.340	−0.466	−0.325	−0.362	−0.533
	[0.084]***	[0.080]***	[0.081]***	[0.106]***	[0.097]***	[0.091]***	[0.093]***	[0.141]***
Δ Argentina's tariffs w.r.t. world								
Outputs			0.207	0.176	−0.003			
			[0.423]	[0.409]	[0.340]			
Inputs				1.126	1.543			
				[0.886]	[0.693]**			
Δ Arg.'s tariffs w.r.t. Brazil								
Outputs						−0.507	−0.780	−0.269
						[0.331]	[0.300]**	[0.344]
Inputs							1.215	0.702
							[0.599]**	[0.524]
Industry-level controls					yes			yes
Firm-level controls		yes	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes
Observations	1,380	1,380	1,380	1,348	1,348	1,374	1,342	1,342
R^2	0.03	0.04	0.04	0.04	0.05	0.04	0.04	0.05
Panel B: Full sample. Dependent variable: export status in 1996								
Δ Brazil's tariffs	−0.291	−0.285	−0.278	−0.203	−0.323	−0.262	−0.281	−0.490
	[0.071]***	[0.077]***	[0.074]***	[0.084]**	[0.101]***	[0.101]**	[0.111]**	[0.144]***
Export status in 1992	0.642	0.543	0.543	0.544	0.546	0.542	0.545	0.544
	[0.023]***	[0.028]***	[0.028]***	[0.028]***	[0.028]***	[0.028]***	[0.029]***	[0.028]***
R^2	0.46	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Panel C: Sample of nonexporters in 1992. Dependent variable: export status in 1996								
Δ Brazil's tariffs	−0.411	−0.446	−0.457	−0.294	−0.447	−0.330	−0.357	−0.604
	[0.108]***	[0.124]***	[0.122]***	[0.123]**	[0.138]***	[0.150]**	[0.175]**	[0.202]***
Observations	797	797	797	781	781	797	781	781
R^2	0.04	0.16	0.16	0.16	0.17	0.16	0.16	0.17

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Firm-level controls include employment measured in efficiency units, sales per worker, and skill intensity, all measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States. In panel B remaining controls and number of observations are the same as in the corresponding column in panel A. In panel C controls are the same as in the corresponding column in panel A.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 4—Entry in the Export Market by Quartile of the Firm Size Distribution

	Full sample						Sample of nonexporters in 1992		
	Change in export status 1996–1992			Export status in 1996			Export status in 1996		
Dependent variable indicated in columns	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Δ Brazil's tariffs									
× first size quartile	−0.331	−0.388	−0.442	−0.125	−0.159	−0.293	−0.358	−0.372	−0.436
	[0.185]*	[0.188]**	[0.198]**	[0.170]	[0.179]	[0.190]	[0.182]*	[0.202]*	[0.226]*
× second size quartile	−0.327	−0.367	−0.412	−0.146	−0.170	−0.306	−0.261	−0.258	−0.318
	[0.146]**	[0.195]*	[0.219]*	[0.175]	[0.212]	[0.228]	[0.230]	[0.260]	[0.278]
× third size quartile	−0.722	−0.784	−0.832	−0.541	−0.576	−0.702	−0.774	−0.720	−0.782
	[0.166]***	[0.151]***	[0.203]***	[0.155]***	[0.152]***	[0.199]***	[0.242]***	[0.239]***	[0.287]***
× fourth size quartile	−0.356	−0.429	−0.483	−0.286	−0.339	−0.474	−0.276	−0.323	−0.377
	[0.175]**	[0.179]**	[0.204]**	[0.119]**	[0.146]**	[0.162]***	[0.342]	[0.390]	[0.379]
Controls									
Export status in 1992				0.553	0.558	0.557			
				[0.027]***	[0.027]***	[0.028]***			
Δ Arg.'s tariffs									
w.r.t. world		yes			yes			yes	
w.r.t. Brazil			yes			yes			yes
Industry-level controls		yes	yes		yes	yes		yes	yes
Firm-level controls	yes	yes	yes	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	1,380	1,348	1,342	1,380	1,348	1,342	797	781	781
R^2	0.03	0.05	0.04	0.50	0.50	0.50	0.15	0.16	0.16

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Controls for changes in Argentina's tariffs with respect to the world and Brazil include both output and input tariffs. Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States. Firm-level controls include dummies for the second, third, and fourth quartile of the firm-size distribution in the initial year (1992).

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.6 Table 4: Export entry by size quartile

Read. The third quartile stands out. The baseline coefficient is −0.722 in first differences, −0.541 in levels, and −0.774 in the 1992-nonexporter sample. These translate into about 13–19 percentage

points higher export entry for third-quartile firms when Brazil's tariffs fall by the average 24 percentage points.

Economic message. The strongest entry effect is not at the bottom or the top. It is in the upper-middle of the size distribution, exactly where the model says the export cutoff should bind.

5.7 Table 5: Average technology-spending response

Read. The baseline coefficient on Δ Brazil tariffs in the $\Delta \log ST$ regression is -1.079, and it remains around -1.05 to -1.45 across the robustness specifications. The average tariff cut implies about a 0.24 log-point increase in technology spending.

Economic message. Export opportunities do not just change who exports. They change how firms produce.

Table 5—Technology Adoption

Dependent variable: change in log (spending on technology)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ Brazil's tariffs	−1.079	−1.077	−1.051	−1.079	−0.939	−1.437	−1.435	−1.449
	[0.350]***	[0.345]***	[0.325]***	[0.340]***	[0.383]**	[0.447]***	[0.483]***	[0.643]**
Δ Argentina's tariffs w.r.t. world								
Outputs			0.556	0.599	0.629			
			[1.116]	[1.147]	[1.186]			
Inputs				−0.762	−0.897			
				[3.211]	[3.040]			
Δ Argentina's tariffs w.r.t. Brazil								
Outputs						2.051	2.254	2.538
						[1.322]	[1.367]	[1.881]
Inputs							−0.361	−0.739
							[2.785]	[3.034]
Industry-level controls					yes			yes
Firm-level controls		yes	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes
Observations	894	894	894	872	872	892	870	870
R^2	0.03	0.05	0.05	0.05	0.06	0.05	0.05	0.06

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Firm-level controls include employment measured in efficiency units, sales per worker, and skill intensity, all measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 6—Product and Process Innovation

Dependent variable indicated in columns	Product and process innovation			Product innovation			Production process innovation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Δ Brazil's tariffs	−0.236	−0.299	−0.400	−0.293	−0.346	−0.438	−0.183	−0.261	−0.357
	[0.104]**	[0.090]***	[0.117]***	[0.116]**	[0.098]***	[0.116]***	[0.100]*	[0.093]***	[0.127]***
Δ Arg.'s tariffs w.r.t. world									
Outputs		−0.191			−0.184			−0.213	
		[0.271]			[0.284]			[0.295]	
Inputs		0.241			0.086			0.360	
		[0.618]			[0.740]			[0.574]	
Δ Arg.'s tariffs w.r.t. Brazil									
Outputs			0.530			0.626			0.341
			[0.308]*			[0.335]*			[0.326]
Inputs			−0.263			−0.584			0.187
			[0.554]			[0.578]			[0.586]
Industry-level controls		yes	yes		yes	yes		yes	yes
Firm-level controls	yes	yes	yes	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes
Observations	1,301	1,269	1,263	1,312	1,280	1,274	1,319	1,287	1,281
R^2	0.24	0.25	0.26	0.22	0.24	0.24	0.22	0.23	0.23

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Firm-level controls include employment measured in efficiency units, sales per worker, and skill intensity, all measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.8 Table 6: Binary innovation measures

Read. Using the innovation index instead of technology spending, the coefficient on Brazil's tariff change ranges from about -0.299 to -0.400 for the aggregate index, with similarly negative effects for product and process innovation separately. Bustos interprets the baseline coefficient as an increase of about 0.07 in the innovation index, or roughly 19% of its mean.

Economic message. The technology result is not just more spending on the same activities. It looks like real changes in innovation behavior.

5.9 Table 7: Technology adoption by size quartile

Read. Again the third quartile is the standout group. For technology spending, the coefficient is -2.106 in the baseline. For the overall innovation index it is -0.359. The other quartiles have smaller and generally less precise coefficients.

Economic message. This is the paper's sharpest success. Trade liberalization induced technology upgrading mainly for firms near the adoption margin, not for firms already far below or above it.

Table 7—Technology Adoption by Quartile of the Firm Size Distribution

Dependent variable indicated in columns	Change in spending on technology			Product and process innovation		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Brazil's tariffs						
× First size quartile	−0.872	−0.725	−1.235	−0.041	−0.076	−0.165
	[0.604]	[0.570]	[0.755]	[0.116]	[0.113]	[0.143]
× Second size quartile	−0.846	−0.662	−1.171	−0.199	−0.227	−0.326
	[0.569]	[0.628]	[0.828]	[0.149]	[0.145]	[0.163]**
× Third size quartile	−2.106	−1.927	−2.424	−0.339	−0.403	−0.465
	[0.609]***	[0.627]***	[0.886]***	[0.133]***	[0.146]***	[0.171]***
× Fourth size quartile	−0.372	−0.146	−0.648	−0.190	−0.229	−0.319
	[0.534]	[0.563]	[0.773]	[0.130]	[0.132]*	[0.154]**
Controls						
Δ Arg.'s tariffs w.r.t. world		yes			yes	
Δ Arg.'s tariffs w.r.t. Brazil			yes			yes
Industry-level controls		yes	yes		yes	yes
Firm-level controls	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes
Observations	894	872	870	1,301	1,269	1,263
R^2	0.05	0.06	0.06	0.20	0.20	0.20

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Controls for changes in Argentina's tariffs with respect to the world and Brazil include both output and input tariffs. Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States. Firm-level controls include dummies for the second, third, and fourth quartile of the firm-size distribution in the initial year (1992).
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 8—Technology Adoption by Initial Export Status

Dependent variable: change in log (spending on technology)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Sample of nonexporters in 1992								
Δ Brazil's tariffs	−1.073	−1.256	−1.304	−1.101	−0.972	−1.586	−1.869	−1.788
	[0.520]**	[0.513]**	[0.498]**	[0.435]**	[0.459]**	[0.682]**	[0.763]**	[0.947]*
Panel B. Sample of exporters in 1992								
Δ Brazil's tariffs	−1.116	−1.043	−0.979	−1.191	−1.052	−1.348	−1.015	−1.153
	[1.382]***	[0.386]***	[0.396]**	[0.403]***	[0.528]**	[0.484]***	[0.555]*	[0.786]
Controls								
Δ Arg.'s tariffs w.r.t. world								
Outputs			yes	yes	yes			
Inputs				yes	yes			
Δ Arg.'s tariffs w.r.t. Brazil								
Outputs						yes	yes	yes
Inputs							yes	yes
Industry-level controls					yes			yes
Firm-level controls		yes	yes	yes	yes	yes	yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Number of observations in panel A is 417, or 407 when input tariffs are included as controls. Number of observations in panel B is 477 in columns 1 to 3, 465 in columns 4 and 6, 475 in column 6, and 463 in columns 7 and 8. Δ denotes a change in a variable during the period 1992–1996. Firm-level controls include employment measured in efficiency units, sales per worker, and skill intensity, all measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 9—Product and Process Innovation by Initial Export Status

Dependent variable indicated in columns	Product and process innovation			Product innovation			Production process innovation		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Sample of nonexporters in 1992									
Δ Brazil's tariffs	−0.179	−0.279	−0.368	−0.220	−0.315	−0.403	−0.145	−0.252	−0.334
	[0.117]	[0.098]***	[0.126]***	[0.126]*	[0.100]***	[0.126]***	[0.115]	[0.097]**	[0.138]**
Panel B. Sample of exporters in 1992									
Δ Brazil's tariffs	−0.324	−0.331	−0.413	−0.403	−0.393	−0.455	−0.247	−0.277	−0.363
	[0.130]**	[0.140]**	[0.184]**	[0.159]**	[0.151]**	[0.206]**	[0.122]**	[0.145]*	[0.181]**
Controls									
Δ Arg.'s tariffs w.r.t. world		yes			yes			yes	
Δ Arg.'s tariffs w.r.t. Brazil			yes			yes			yes
Firm-level controls	yes	yes	yes	yes	yes	yes	yes	yes	yes
Industry-level controls		yes	yes		yes	yes		yes	yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Number of observations in panel A is 741, in column 1, 725 in columns 2 and 3, 747 in column 4, 731 in columns 5 and 6, 753 in column 7, and 737 in columns 8 and 9. Δ denotes a change in a variable during the period 1992–1996. Controls for changes in Argentina's tariffs with respect to the world and Brazil include both output and input tariffs. Firm-level controls include employment measured in efficiency units, sales per worker, and skill intensity, all measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.10 Tables 8 and 9: Mechanism by initial export status

Read. In Table 8, both nonexporters and exporters in 1992 show negative and usually significant coefficients in the $\Delta \log ST$ regressions. In Table 9, the innovation indexes tell the same story for both groups.

Economic message. This is important because it favors the revenue mechanism over a pure "learning-by-exporting" story. Even firms already exporting upgrade when export revenues rise.

5.11 Table 10: Industry-level exports to Brazil

Read. The industry-level export regressions produce large negative coefficients on Brazil's tariff changes, roughly between -2.8 and -4.2 in OLS and IV. The IV uses 1991 tariffs as an instrument for the tariff change and has a very strong first stage.

Economic message. The paper's mechanism needs Brazilian tariff cuts to raise actual exports to Brazil, and Table 10 supports exactly that.

Table 10—Export Sales to Brazil

Dependent variable: change in (log) export sales 1992–1996	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. OLS						
Δ Brazil's tariffs	−2.836	−3.598	−2.402	−3.113	−2.621	−2.291
	[1.560]*	[1.346]***	[1.622]	[1.325]**	[1.221]**	[1.325]*
R^2	0.39	0.45	0.45	0.07	0.19	0.17
Panel B. IV						
Δ Brazil's tariffs	−3.513	−4.226	−3.129	−3.911	−3.326	−3.242
	[1.448]**	[1.274]***	[1.411]**	[1.339]***	[1.273]***	[1.341]**
R^2	0.39	0.45	0.45	0.07	0.19	0.16
Panel C. First stage						
Brazil's tariffs in 1991	−0.774	−0.764	−0.747	−0.763	−0.767	−0.744
	[0.035]***	[0.040]***	[0.038]***	[0.026]***	[0.029]***	[0.032]***
R^2	0.95	0.95	0.95	0.94	0.94	0.94
F-test on excluded instrument	480.12	366.19	378.54	867.08	685.09	548.69
Controls						
Δ Arg.'s tariffs						
with respect to world		yes			yes	
with respect to Brazil			yes			yes
Industry-level controls		yes	yes		yes	yes
2-digit ISIC industry dummies	yes	yes	yes			
Observations	100	97	96	100	97	96
R^2	0.39	0.45	0.45	0.07	0.19	0.16

Notes: Robust standard errors in brackets. Δ denotes a change in a variable during the period 1992–1996. Controls for changes in Argentina's tariffs with respect to the world and Brazil include both output and input tariffs. Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 11—Domestic Sales, Effect of Argentina's Output Tariff Reductions w.r.t Brazil

Dependent variable: change in log (domestic sales) 1996–1992	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ Arg.'s tariffs w.r.t. Brazil							
Outputs	1.315	1.465	1.022	1.278	1.022	0.662	0.619
	[0.545]**	[0.573]**	[0.541]*	[0.545]**	[0.538]*	[0.585]	[0.701]
Inputs		−1.010	−0.591	−0.579		−1.161	−1.047
		[1.556]	[1.417]	[1.351]		[1.285]	[1.241]
Δ Brazil's tariffs					0.191	0.306	0.438
					[0.256]	[0.211]	[0.271]
Firm-level controls			yes			yes	
Industry-level controls				yes			yes
2-digit ISIC industry dummies	yes	yes	yes	yes	yes	yes	yes
Observations	1,371	1,339	1,339	1,339	1,371	1,339	1,339
R^2	0.06	0.06	0.08	0.06	0.06	0.08	0.06

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996. Firm-level controls include employment measured in efficiency units and skill intensity, both measured in the initial year (1992). Industry-level controls include demand elasticity, skill intensity, and capital intensity of the 4-digit ISIC industry in the United States.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table 12—Domestic Sales, Effect of Other Tariff Reductions

Dependent variable: change in log (domestic sales) 1996–1992	(1)	(2)	(3)	(4)	(5)
ΔBrazil's tariffs	0.368				
	[0.246]				
Δ Arg.'s tariffs w.r.t. Brazil					
Inputs		0.579			
		[1.442]			
Δ Arg.'s tariffs w.r.t. the world					
Outputs			−1.171		−1.425
			[0.786]		[0.788]*
Inputs				2.358	2.551
				[1.592]	[1.635]
2-digit ISIC industry dummies	yes	yes	yes	yes	yes
Observations	1,377	1,345	1,377	1,345	1,345
R^2	0.05	0.05	0.05	0.05	0.06

Notes: Standard errors are clustered at the 4-digit ISIC industry level. Δ denotes a change in a variable during the period 1992–1996.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.12 Tables 11 and 12: Domestic sales

Read. Lower Argentine output tariffs against Brazil are associated with lower domestic sales. The baseline coefficient in Table 11 is 1.315, implying that the average 13 percentage-point tariff reduction reduced domestic sales by about 0.17 log points. But this result weakens with additional controls. Table 12 shows that other tariff changes have little effect on domestic sales.

Economic message. The domestic-revenue side of the mechanism is suggestive rather than definitive. Still, the evidence points in the expected direction: liberalization with Brazil made

domestic competition tougher.

6 Short summary

Bustos studies whether trade liberalization can induce firms to upgrade technology, not just reallocate output toward firms that were already better. She embeds fixed-cost technology adoption into a heterogeneous-firm trade model and then uses the MERCOSUR tariff cuts between Brazil and Argentina as quasi-experimental variation in export opportunities. The empirical results show that larger Brazilian tariff cuts increased the probability of export entry and increased technology spending and innovation, with the effects concentrated in the upper-middle part of the firm-size distribution. The fact that already-exporting firms also upgraded technology supports the paper's core mechanism: better export opportunities raise revenues enough to make advanced technology adoption profitable.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that falling partner-country tariffs can induce firms to adopt better technology. The main message is not merely that exporters are different from nonexporters, but that trade policy can *change* firms' technology choices.

7.2 Economic intuition

Technology adoption has a fixed cost. Exporting raises revenues. When revenues rise enough, some firms cross a profitability threshold and find it worthwhile to upgrade. The firms most likely to do so are neither the weakest nor the strongest, but those in the upper-middle range of productivity that are close to the adoption margin.

7.3 Takeaway

This paper demonstrates that trade liberalization can work through an endogenous technology-upgrading channel. That insight is important for thinking about productivity, development, and the broader gains from opening export markets.